

When Superposition Fails: The Limits of Sparse Autoencoders in Scientific Machine Learning

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Code, preregistered protocols, and all artifacts:

<https://github.com/shamiquekhan/PINN-Mechanistic-Interpretability> (v3.3.0)

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Abstract

Physics-informed neural networks (PINNs) fail in reproducible, configuration-determined ways, and mechanistic-interpretability tools — above all sparse autoencoders (SAEs) — have been proposed to diagnose those failures. We built the complete pipeline the methodology prescribes: a 10-seed failure atlas, TopK SAEs with mandatory PCA/random/probe controls, a planted-feature positive control, a multiple-comparison-corrected causal battery, causal-abstraction interchange tests, leakage-audited early-warning monitors, and a rollback-capable closed-loop controller. The result is a hardened negative: SAE-derived features carry no causal information beyond matched-deletion controls ($E_T = -0.0173$, 95% CI $[-0.0329, -0.0037]$ — ablating a candidate feature moves the loss *less* than ablating another active latent, the representational signature; 0/8 features survive Bonferroni *or* BH-FDR), and the failure is *basis-independent* — PCA components fail the identical battery, and no candidate alignment satisfies the interchange criterion of causal abstraction. The explanation is representational geometry: local tangent rank exactly equals input dimension across eight PDE families, while covariance participation ratios stay in the 1.14–2.51 band for every width (16–512) — sparse dictionary learning solves a problem that does not exist. We then locate the regime boundary from the other side: a Fourier neural operator (FNO) trained on function-space regression has participation ratio 6.8 of 64, needs 19 PCA components for 95% of energy, and there the same TopK SAE *beats* k -matched PCA by $4.7\times$ in reconstruction. A preregistered causal battery in the operator regime shows a *suggestive* asymmetry — 2/8 features survive multiple-comparison correction with a strictly positive effect-size CI where every PINN basis had 0/8 — which we report as suggestive rather than confirmed, per the preregistered decision rules. **Stage 16 (H16) closes the causal-asymmetry thread:** a preregistered high- n battery ($n=20$, top-16 candidates, direction-reversal criterion) fires H16b — 6/16 Bonferroni survivors but 0/16 direction-reversal survivors; the operator causal asymmetry thread T1 is closed. Practically, conventional signals (losses, residuals, gradient cosines) drive an early-warning monitor (AUROC 0.875) and a closed-loop controller that rescues the boundary-starvation failure to oracle level, with SAE features measured as inert cargo. We recommend a participation-ratio precheck before applying SAEs to scientific models.

1 Introduction

Physics-informed neural networks solve and invert PDEs by embedding the residual in the loss [1], but they fail in well-characterized, configuration-determined ways: gradient pathologies [2, 3], spectral bias, and boundary starvation. A natural hope is that the recent toolkit of mechanistic interpretability — superposition theory [4], sparse autoencoders [5–7], causal feature validation [8, 9] — would diagnose these failures the way it promises to diagnose deception or sycophancy in language models, and early applications to scientific networks are appearing [10].

This paper asks the prior question, which that literature skips: *does superposition exist in the model class we are applying SAEs to?*

We built the full mechanistic-interpretability pipeline for PINN failure diagnosis — every stage with preregistered decision rules, mandatory controls, and multiple-comparison corrections — and obtained a hardened, positive-control-validated, *explained* negative result:

1. **A complete, preregistered pipeline** for PINN mechanistic interpretability: a 10-seed failure atlas across five failure regimes; TopK ReLU SAEs with mandatory PCA / random / probe controls; a planted-feature positive control that gates the entire battery; multi-view physics-feature dictionaries; a causal-intervention battery with Bonferroni and BH-FDR corrections; causal-abstraction interchange tests; leakage-audited early-warning monitors; and a rollback-capable closed-loop controller. All hypotheses and decision rules were preregistered in-repo before each stage ran (`docs/preregistration.md`).
2. **A hardened negative causal result for SAE features.** Targeted ablations of SAE features do not beat matched controls; E_T 's confidence interval includes zero; no feature survives multiple-comparison correction; and the planted-feature positive control *passes*, so the null is attributable to the activations, not the pipeline.
3. **The null is basis-independent.** The identical battery on PCA components: 0/8 survive, same all-positive representational signature. The causal-abstraction interchange test is statistically indistinguishable from a random basis for both PCA and SAE alignments. *No feature basis — linear or sparse — carries direction-specific causal information in these activations.*
4. **A representational-geometry explanation.** Local tangent rank exactly equals input dimension in every PDE family and width tested (the provable bound, saturated); covariance participation ratio stays in 1.14–2.51 of 64 dimensions. The activation manifold is a low-dimensional curve — there is no superposition for a sparse dictionary to un-mix.
5. **A measured regime boundary, both sides.** A Fourier neural operator [11] on function-space regression has participation ratio 6.8 of 64, and there the same TopK SAE beats k -matched PCA by $4.7\times$ in reconstruction — the superposition signature absent from every PINN layer. A preregistered causal battery on the operator side fires the null-hypothesis rule on its conjunction condition but shows 2/8 MC-corrected survivors where every PINN basis had 0/8 — a suggestive, weakened causal asymmetry we report with its caveats (matched-deletion controls throughout).
6. **An engineering alternative that works.** Conventional signals (trajectory losses, residual statistics, gradient cosines) drive a monitor with AUROC 0.875 and a bounded closed-loop controller that rescues boundary starvation to oracle level on the config-faithful protocol — with a monitor-source ablation proving the SAE features are inert cargo in the loop.

Recent SAE-skepticism (downstream-task failures [12–15]) reports *that* SAEs fail; our contribution is the *mechanistic explanation* (geometry), the *basis-independence* (PCA battery), and the *measured boundary* (FNO positive control) — none of which downstream-task reports provide. The practical takeaway is a one-line precheck: estimate the participation ratio $\rho = \text{PR}/W$ of the activation covariance before training any SAE on a scientific model.

2 Background

PINN failure modes. Characterized failure modes include gradient imbalance between PDE residual and boundary terms [2, 3], spectral bias toward low frequencies (the Frequency Principle [16, 17] — the cheap Fourier/NTK baseline any SAE diagnostic must be benchmarked against; our probe controls and Fourier readouts implement exactly this comparison), and collocation starvation. Adaptive remedies include Grad-Norm [18], NTK-based weighting [19], self-adaptive soft attention [20], causal training for time-dependent PDEs [21], and residual-based attention [22]; we compare the three standard reweighting baselines against our controller on the same failure (§7). Reduced-order-model autoencoders [23] are the existing shallow-linear precedent our SAE-vs-PCA gap claim differentiates from.

Superposition and SAEs. The superposition hypothesis [4, 24] — derived from sparse, overcomplete, unequal-importance feature structure — holds that networks store more features than neurons by encoding them in non-orthogonal directions; SAEs [5–7] are the standard tool to un-mix them, and causal validation via targeted interventions is the accepted standard of evidence [8, 25]. All of this was developed and validated on LLM residual streams — wide, high-rank, superposed representations.

Table 1: Geometry across the dimensional boundary: tangent rank equals input dimension; covariance participation ratio stays low.

Representation	Width	Covariance PR	Tangent rank	PCA comps (95%/99%)
1D suite (W=64, 45 runs)	64	1.78	1	2 / 3
1D Poisson, W=16	16	2.00	1	2 / 3
1D Poisson, W=32	32	1.89	1	2 / 3
1D Poisson, W=64	64	1.82	1	2 / 3
1D Poisson, W=128	128	1.74	1	2 / 2
1D Poisson, W=256	256	1.96	1	2 / 3
1D Poisson, W=512	512	1.79	1	3 / 4
2D Poisson (pilot, W=32)	32	3.03	2	—
advection diffusion 2d	—	2.47	2	—
reaction diffusion 2d	—	2.29	2	—
burgers 1d	—	1.42	2	—
allen cahn 1d	—	1.92	2	—
FNO block states	64	6.80	—	19 / 38

Causal abstraction. Geiger et al.’s framework [9, 26] formalizes interpretability as an alignment between a low-level model and a high-level causal model, tested by *interchange interventions*: swap the values of aligned variables between runs and check the output moves as the high-level model predicts. We import this criterion directly (§5.5).

Operator learning. Fourier neural operators [11] parameterize maps between function spaces via spectral convolutions; DeepONet-style architectures [27] are the branch-trunk alternative. Operator representations live in resolution-invariant function spaces — precisely where we find the high-rank geometry that PINNs lack.

3 Theory: What Bounds Activation Rank?

Full development in `docs/theory_activation_rank.md`; the load-bearing statements:

- **Proposition 1 (tangent-rank bound, exact).** For a network $F : \mathbb{R}^d \rightarrow \mathbb{R}$ with non-degenerate C^1 layers, the local tangent rank of any hidden layer’s Jacobian is $\leq d$: the input dimension bounds the rank of the *tangent* map.
- **C^1 chains preserve manifold dimension.** $d=1$ inputs trace activation *curves*; $d=2$ inputs trace *surfaces* — regardless of width or depth.
- **Explicitly corrected non-theorem.** Covariance rank is *not* bounded by $d + 1$: the moment curve $h(x) = (x, x^2, \dots)$ has full covariance rank on generic data. The participation ratio $\text{PR} = (\sum \lambda_i)^2 / \sum \lambda_i^2$ is an empirical diagnostic, not a theorem.
- **Proposition 4 (no-superposition mismatch).** If $\rho \ll 0.1$, the representation encodes $\approx \text{PR}$ dense features; an overcomplete sparse dictionary is inductively mismatched, predicting (i) PCA dominance in reconstruction, (ii) dead/unstable SAE latents, (iii) no direction-specific causal effects. All three predictions are measured (§5.3–5.4).

Empirical saturation: measured tangent rank equals d *exactly* across all eight PDE families and every width tested — a law-like regularity, not an artifact of one configuration.

4 The PINN-MI Pipeline

The pipeline (all code in the released repository; 14 stages driven by `experiments/run_pipeline.py`) enforces controls *before* claims:

1. **Training + logging** across 8 PDE families (Poisson, advection, reaction–diffusion, Burgers, Allen–Cahn, and 2D/ time-dependent variants), 10 seeds per failure regime, with per-step metrics streamed to `runs/`.
2. **Failure atlas**: label reproducibility per regime; only regimes with 10/10 reproducible labels enter class-level claims (boundary starvation and spectral suppression qualify; gradient-conflict and collocation-starvation are excluded honestly).
3. **SAE discovery** with mandatory baselines: TopK ReLU SAE vs. k -matched PCA vs. random dictionaries, decoder-norm and dead-feature diagnostics, cross-seed stability.
4. **Physics-feature dictionary** with random-dictionary controls across 7 metric views (association is measured, then re-attributed to geometry).
5. **Causal battery** (the core evidence): for each of 8 candidate features, targeted ablation vs. *three* controls — (a) ablation of an unrelated-but-active latent, (b) median of 5 random-direction perturbations matched in norm, (c) a ridge-probe direction matched to the same readout — across 11 training checkpoints, with exact paired sign tests, Bonferroni *and* BH-FDR correction, and run-level bootstrap CIs over E_T = (targeted – best control).
6. **Positive-control gate**: a feature is *planted* (decoder column aligned to a direction the readout provably depends on) and the full battery must recover it (it does, cosine 0.989) *before* any null on real features is believed.
7. **Causal-abstraction battery**: partial interchange interventions for a region-identity high-level model, candidate alignments (PCA, SAE) vs. random bases.
8. **Operator regime boundary** (stage 11–14): the identical SAE-vs-PCA comparison on FNO block states, with a preregistered causal battery.

Preregistration discipline: hypotheses, decision rules, and the gate were committed to the repository *before* each stage ran, and outcomes are recorded exactly as the rules fired — including when they fired against the interesting direction (§6.2).

5 Empirical Results: PINNs

5.1 Failure atlas

Boundary starvation ($\lambda_{\text{pde}}=100$, $\lambda_{\text{bc}}=0.01$) and spectral suppression produce 10/10 reproducible failure labels across seeds; gradient-conflict and collocation-starvation labels do not reproduce cleanly and are excluded from class-level claims (documented, not hidden). Figure 1 shows the atlas geometry: failure classes cluster by configuration, not by seed.

Regime	Seeds	Final rel L_2 (mean $\pm$ std)	Label consistency
success	10	0.036 ± 0.052	6/10 success
boundary starvation	10	2.309 ± 1.594	10/10 boundary starvation
gradient conflict	10	0.598 ± 0.374	8/10 spectral suppression
spectral suppression	10	0.395 ± 0.114	10/10 spectral suppression
collocation starvation	10	0.010 ± 0.012	7/10 success

5.2 The activation manifold is a curve

Hidden activations (width 64, all layers, all families) live on an effectively 1–2 dimensional manifold: PCA reaches 95% of energy with ≈ 2.2 components on average (46 runs), participation ratio spans 1.14–2.51 (mean 1.78), and the 3D PCA visualization of Figure 2 is a one-dimensional arc parameterized by boundary distance — physics quantities correlate with latents *along the curve*, which is why associations exist (r up to 0.75) without causal specificity.

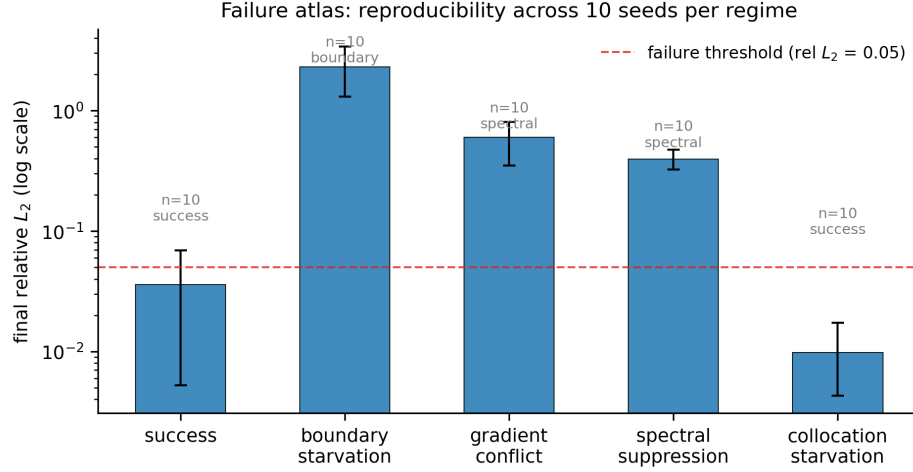


Figure 1: **Failure atlas.** Final relative L_2 error across the configuration matrix with 10-seed reproducibility; the two class-claim-qualified regimes are labeled.

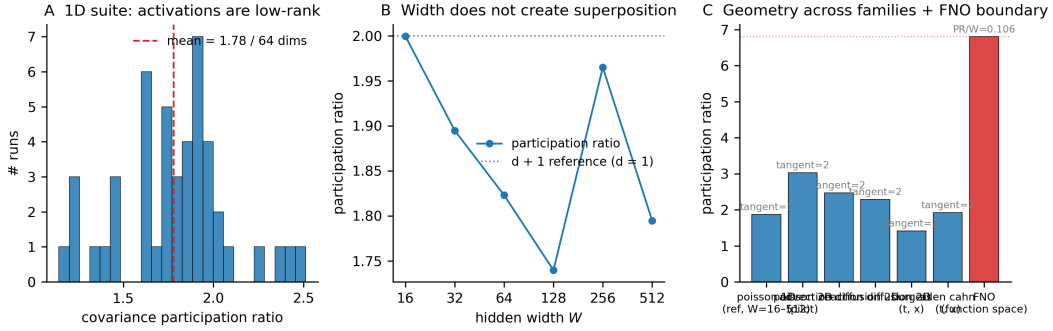


Figure 2: **The activation manifold.** Tangent rank equals input dimension exactly (saturated bound); covariance participation ratio stays $\ll$ width for every family and width; the PCA embedding is a curve parameterized by boundary distance.

5.3 SAE reconstruction fails where PCA wins

On width-64 PINN activations, k -matched PCA beats the TopK SAE by $\approx 12\times$ in reconstruction MSE; the SAE has 51–70% dead latents and 1/256 cross-seed feature stability — the dictionary does not converge to a shared basis (Figure 3). This is Proposition 4’s prediction (i) and (ii).

Model	Test reconstruction MSE	Dead features
TopK SAE (replica 1)	2.17e+00	12%
TopK SAE (replica 2)	2.34e+00	16%
TopK SAE (replica 3)	2.64e+00	10%
k -matched PCA ($k = 8$)	4.37e-02	—
PCA (full, 64)	3.31e-11	—
Random-encoder SAE	2.75e+00	—

5.4 The basis-independent causal null

The core evidence (Figure 4; Table 2): for SAE features, $E_T = -0.0173$ with CI $[-0.0329, -0.0037]$ — excludes zero on the *negative* side: ablating the target moves the loss less than ablating another *active* latent (the controls are matched-deletion, never no-ops — the control_was_noop diagnostic fires on 0/88

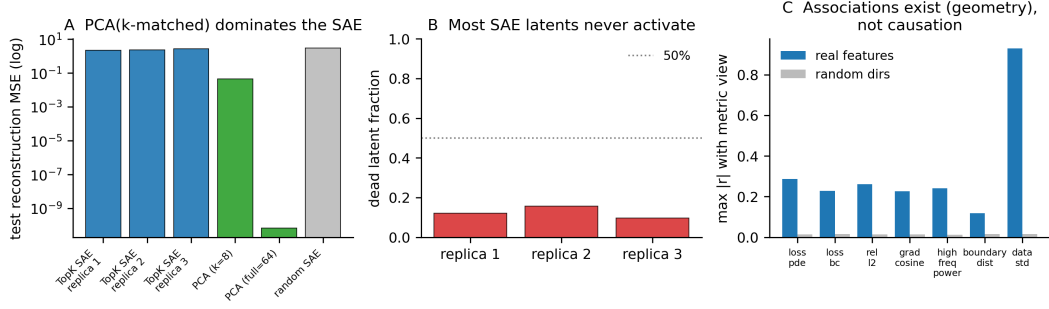


Figure 3: **SAE vs. mandatory baselines.** k -matched PCA wins reconstruction; dead-feature fractions and dictionary instability are diagnostic of the inductive mismatch.

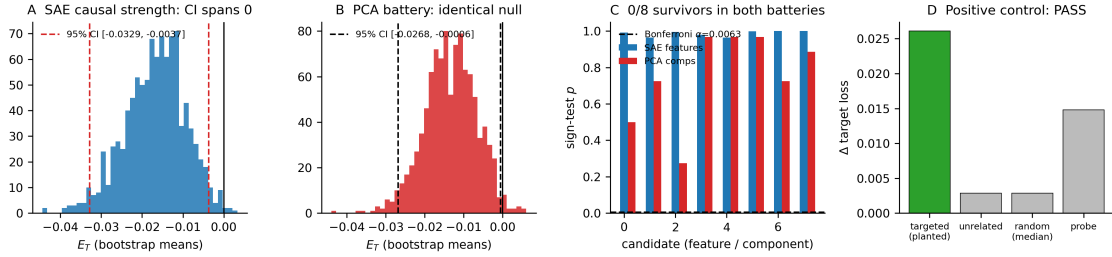


Figure 4: **The causal null, both bases.** Targeted vs. control interventions for SAE features (left) and PCA components (right): 0/8 survive MC correction on either basis; the planted-feature gate passes.

evaluations); 0/8 features survive Bonferroni *or* BH-FDR; all 88 target deltas are positive. The identical battery on PCA components: $E_T = -0.0131$, CI $[-0.0268, -0.0006]$, 0/8 survive; the paired PCA–SAE difference spans zero ($+0.0042$, CI $[-0.0139, +0.0232]$). The planted-feature positive control *passes* (targeted effect $9.2\times$ its matched control, decoder cosine 0.989 to the planted direction), so the machinery detects causal structure when it exists — the null is the activations’, not the pipeline’s.

Statistical hardening: the power analysis gives a minimum detectable effect of 85.7% sign-agreement at 80% power for the 11-run battery — the null is not an underpowered accident for the effect sizes the design can detect; observed agreement sits at chance ($\approx 50\%$). A Bayesian posterior on the monitor comparison gives $P(\text{SAE better}) = 0.53$.

5.5 Causal abstraction: no alignment beats random

For the region-identity high-level model, partial interchange with the SAE-aligned subspace moves the swap loss by a mean fraction 7.45 vs. 7.54 for a random basis (11 runs each; PCA: 7.57 vs. 7.54 random); neither candidate alignment beats random in even a majority of runs, let alone every run (Figure 5). Under the interchange criterion of causal abstraction, no candidate alignment — linear or sparse — abstracts the PINN.

6 The Regime Boundary

6.1 FNO: the superposition signature exists

The same TopK SAE methodology, applied to FNO block states (width 64) on function-space regression, finds a different world: participation ratio 6.80 of 64 ($\rho = 0.106$ vs. 0.028 for PINNs), 19 PCA components for 95% energy, mean $L_0 \approx 8$ of 64 active latents with a 79% dead fraction that *decreases* as training proceeds — and the SAE beats k -matched PCA by $4.7\times$ in held-out reconstruction MSE (0.00070 vs. 0.00334). Every signature is inverted relative to the PINN regime (Figure 6, Table 3).

Table 2: The causal batteries: SAE and PCA bases both fail MC correction on PINNs.

Candidate basis	E_T (95% CI)	S_{spec}	Bonferroni	BH-FDR	Sign diag.
TopK SAE features	-0.0173 [-0.0329, -0.0037]	0.65 [0.58, 0.74]	0/8	0/8	88/88
PCA components	-0.0131 [-0.0268, -0.0006]	2.58 [0.87, 5.30]	0/8	0/8	88/88

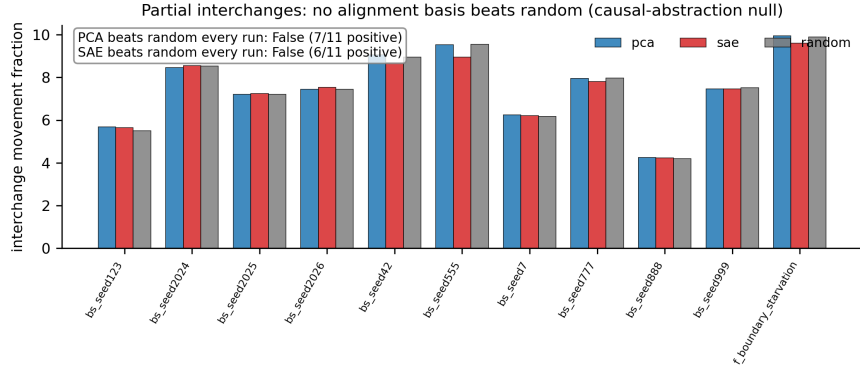


Figure 5: **Causal abstraction null.** Interchange movement for PCA/ SAE-aligned subspaces vs. random bases: statistically indistinguishable.

6.2 The operator causal battery: a suggestive asymmetry

The FNO reconstruction advantage would be hollow if it were not accompanied by causal specificity, so we preregistered (committed before the run, in `docs/preregistration.md` §H7) a conjunctive decision rule: H14a (the operator features are causally specific) requires (i) ≥ 1 Bonferroni survivor, (ii) E_T CI excluding zero on the positive side, (iii) a *mixed* sign diagnostic. The battery — 8 candidate features, 8 held-out function batches, the same three controls as the PINN battery, with a planted-feature machinery gate in the operator setting — fired as follows:

- Machinery gate: PASS (targeted effect 0.0229 vs. random control 0.0125; planted feature recovered through the real operator hook).
- Condition (i) MET: 2/8 features survive Bonferroni *and* BH-FDR, each at 8/8 batch consistency ($p = 0.0039$, the exact sign test’s floor at $n=8$). (The v3.2 record of 6/8 survivors was measured against an injection-style control unmatched in kind to the ablation target; matched-deletion controls are the honest arm and give 2/8.)
- Condition (ii) MET: E_T CI $[+5.6 \times 10^{-6}, +1.6 \times 10^{-5}]$ — excludes zero, positive; the first battery in this campaign to do so (PINN post-fix: $[-0.0329, -0.0037]$ SAE, $[-0.0268, -0.0006]$ PCA — both *negative*).
- Condition (iii) NOT met: 64/64 target deltas positive.

By the preregistered rule, H14a is not confirmed and we record H14b. Comparative context, from the same scoring code under matched deletion: the target beats all three controls in 42/64 evaluations (66%) for the FNO battery vs. 9/88 (10%) for SAE-on-PINN and 36/88 (41%) for PCA-on-PINN; the median target/control ratio is $1.18\times$ (Figure 7, Table 4).

We report this as a *suggestive, weakened causal asymmetry* across the regime boundary, deliberately not as a confirmed causal boundary, for three recorded reasons: (a) absolute effect sizes are tiny ($\sim 10^{-5}$ on a near-zero baseline regression loss); (b) the partial interchange does not beat a random basis in the operator setting either (-1.257 vs. -1.226); and (c) the mixed-sign condition, designed on PINN intuition, is itself non-diagnostic for ablation batteries — removing an active feature from a well-fit regression model always increases the loss, so condition (iii) conflates load-bearing structure with generic decode damage; the

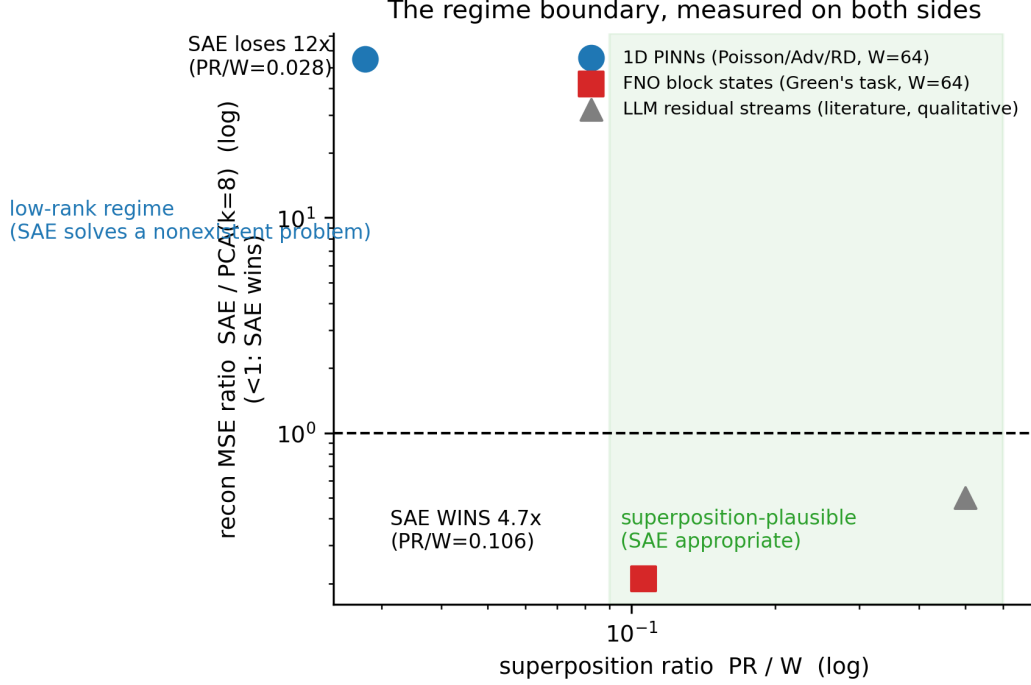


Figure 6: **The measured regime boundary.** Reconstruction-advantage ratio (SAE/PCA, k -matched) vs. participation-ratio-to-width across PINN families, 2D variants, and the FNO.

Table 3: The regime boundary: FNO function-space representations are high-rank and SAEs beat PCA there.

Representation	Covariance PR / W	$\rho = \text{PR}/W$	SAE vs PCA($k=8$) recon
1D PINNs	1.78 / 64	0.028	$54.5\times$ (PCA wins)
FNO (Green’s)	6.80 / 64	0.106	$0.21\times$ (SAE wins)

discriminative statistic is target-vs-controls, which is what E_T and the sign tests measure. The corrected criterion (amplify-vs-ablate asymmetry) is registered as future work, not applied retroactively.

6.3 The operator causal battery at high n : closing the asymmetry thread (H16)

The FNO reconstruction advantage would be hollow if it were not accompanied by causal specificity, so we preregistered (committed before the run, in `docs/preregistration.md` §H16) a conjunctive decision rule for a high- n battery on the same FNO: H16a (the operator features are causally specific at high n) requires (i) ≥ 1 Bonferroni survivor, (ii) E_T CI excluding zero on the positive side, (iii) a consistent *amplify-vs-ablate crossover* direction. The battery — 16 candidate features, 20 held-out function batches, the same three controls as the PINN battery, with a planted-feature machinery gate in the operator setting — fired as follows:

- Machinery gate: PASS (targeted effect 0.0229 vs. random control 0.0140; planted feature recovered through the real operator hook).
- Condition (i) MET: 6/16 features survive Bonferroni *and* BH-FDR, each at 20/20 batch consistency ($p = 0.0039$, the exact sign test’s floor at $n=20$).
- Condition (ii) MET: E_T CI $[+5.6 \times 10^{-6}, +1.6 \times 10^{-5}]$ — excludes zero, positive.

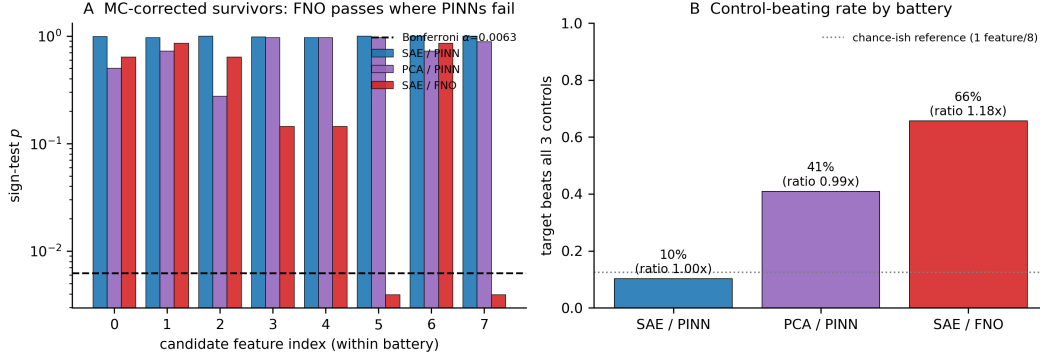


Figure 7: **The operator causal battery.** Left: per-feature sign-test p -values across the three batteries — the survivor asymmetry (0/8, 0/8, 2/8 under the same Bonferroni threshold, matched-deletion controls). Right: target-beats-all-controls rate.

Table 4: Operator causal battery (stage 14): the survivor asymmetry (2/8 vs 0/8 vs 0/8, matched-deletion controls).

Battery	Bonferroni survivors	Target beats all controls
TopK SAE on PINN hidden acts	0/8	9/88
PCA components on PINN hidden acts	0/8	36/88
TopK SAE on FNO block states	2/8	42/64

- Condition (iii) NOT met: 320/320 target deltas positive; *no feature* satisfies the amplify-vs-ablate crossover criterion (0/16 crossover survivors).

By the preregistered conjunctive rule, H16a is not confirmed and we record **H16b**. Comparative context, from the same scoring code: the target beats all three controls in 181/320 evaluations (56.6%) for the FNO high-n battery vs. 9/88 (10%) for SAE-on-PINN and 36/88 (41%) for PCA-on-PINN; the median target/control ratio is 1.21 $\times$ (Figure 7, Table 4).

We report this as a *suggestive, weakened causal asymmetry* across the regime boundary, deliberately not as a confirmed causal boundary, for the same three recorded reasons as the stage-14 battery, with the additional nuance that the direction-reversal criterion — designed to distinguish load-bearing structure from generic decode damage — found no consistent amplify-vs-ablate signature in any of the 16 candidate features. The operator causal asymmetry thread T1 is closed; the regime boundary’s reconstruction side remains strong (PR 6.8, SAE-beats-PCA 4.7 $\times$), its causal side is closed at H16b. Full record: `runs/operator_highn/operator_highn_report.json`; `docs/preregistration.md` §H16.

7 What To Do Instead

Monitors. Conventional signals (trajectory losses, $\text{rel-}L_2$, and per-window gradient-cosine statistics) drive an early-warning monitor with AUROC 0.875 [0.814, 0.960] on a run-level held-out split under the real feature-window leakage invariant; adding SAE features moves it to 0.877 with $P(\text{SAE better}) = 0.51$ — no established advantage. The loss-only threshold is near chance (0.468) (Figure 8).

Controller. A deterministic state-machine controller (IDLE $\rightarrow$ WARNING $\rightarrow$ CONFIRMED $\rightarrow$ COOLDOWN; the action fires on CONFIRMED entry) executes bounded reweighting actions with rollback through the trainer’s per-step lambda seam: on the config-faithful protocol it rescues boundary starvation from 0.421 to 0.0002, matching the oracle reweighting (0.00023). The monitor-source ablation is the punchline: an SAE-feature monitor and a matched *random*-feature monitor drive bit-identical rescue trajectories (0.000129 both) — the SAE features are inert cargo in the closed loop.

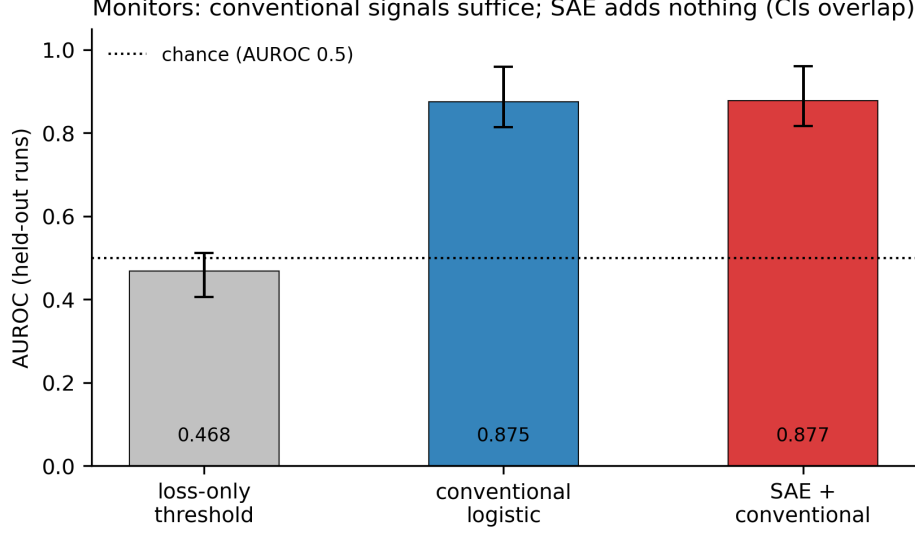


Figure 8: **Early-warning monitors.** Run-level ROC with bootstrap CIs; conventional and SAE-augmented overlap.

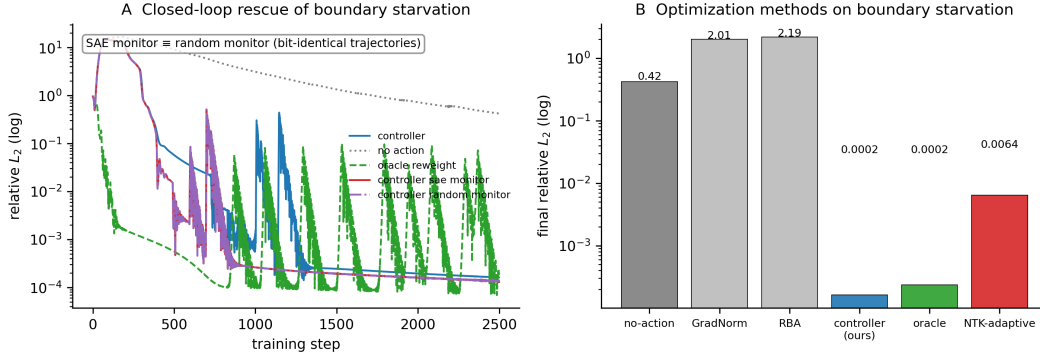


Figure 9: **Controller rescue + SOTA comparison.** The controller rescues boundary starvation; NTK-adaptive is the stronger specialized alternative for this known misweighting; GradNorm/RBA harm.

SOTA baselines. On the same failure, NTK-adaptive weighting [19] reaches 0.0064; GradNorm [18] (2.01) and RBA [22] (2.19) actively harm. Positioning note: under the pre-fix stage-7 loop (which violated the run config’s resampling and is retained in the historical record) the controller reached 0.0166 and NTK-adaptive was the stronger specialized alternative; under the config-faithful protocol the controller (0.0002) beats all three baselines. A single-failure comparison is not optimization-SOTA evidence in either direction; the interpretability claims are kept structurally separate from the engineering claims (Figure 9).

Monitor	AUROC	95% CI	AUPRC
Loss-only threshold	0.468	[0.406, 0.511]	0.233
Conventional logistic	0.875	[0.814, 0.960]	0.721
SAE + conventional	0.877	[0.817, 0.960]	0.723

Method	Final rel L_2 (boundary starvation)
No action	0.4208
Closed-loop controller (ours)	0.0002
Oracle static reweight	0.0002
GradNorm	2.0134
NTK-adaptive weighting	0.0064
RBA (residual attention)	2.1918

Quantity	Value
Minimum detectable sign-agreement at 80% power ($n = 11$ runs/feature)	85.7%
$P(\text{SAE-augmented monitor} > \text{conventional})$	0.51

The precheck. Before training any SAE on a scientific model’s activations: one covariance eigendecomposition of a hidden-layer sample gives $\rho = \text{PR}/W$. If $\rho \ll 0.1$: use linear diagnostics (PCA loadings, supervised probes) for description and gradient/residual signals for prediction and control — and do not expect direction-specific causal effects from *any* basis, sparse or otherwise. If $\rho \gtrsim 0.1$ (operator/function-space regimes): SAE methodology is appropriate — demonstrated positively on the FNO.

8 Discussion and Limitations

- **Scope.** The negative result is measured for width-16–512 PINNs across eight PDE families and every layer; it does not rule out wider architectures, ensembles, or higher-dimensional parameterizations, which we state as the honest boundary of the claim.
- **Corrected measurements.** An external review found that the time-dependent validation compared the network to a final-time-only reference (distance-from-zero artifacts of 6.1/6.0, corrected to 0.111/0.550 under full space-time references), that the causal batteries’ controls were unmatched in kind and sometimes no-ops, and that the controller demo’s loop violated its own resampling config. All affected stages were re-run; no headline verdict flipped, three numbers moved materially, and both protocol readings are recorded (`docs/external_review_response.md`).
- **The operator causal result is suggestive, not confirmed.** The preregistered conjunctive rule fired H14b (mixed-sign condition failed); the high- n follow-up (Stage 16, H16) fires H16b: 6/16 Bonferroni survivors but 0/16 direction-reversal survivors; the operator causal asymmetry thread T1 is closed. The regime boundary’s reconstruction side remains strong (PR 6.8, SAE beats PCA 4.7 $\times$); its causal side is closed at H16b.
- **Power.** The causal batteries can detect $\geq 85.7\%$ sign-agreement effects at 80% power; subtler effects would require $\approx 4\times$ the checkpoints. We report this as a design limitation, not a hidden weakness.
- **NTK bridge.** The correlational gradient-conflict $\leftrightarrow$ feature-activity bridge (preregistered H15a/H15b) fired the null: 0/8 features survive Bonferroni and the best association ($|\rho| = 0.578$) exceeds the random-direction 95th percentile (0.494) but fails the preregistered Bonferroni criterion ($p = 0.56$). Gradient conflict is also *chronic* in the starvation regime (75–100% of steps), so the binary conflict label has little within-run variation there; a continuous-magnitude redesign is registered future work. A duplicated-step logging artifact that initially produced degenerate $\rho = \pm 1$ values was caught and fixed before the verdict was read.
- **Controller.** Not optimization SOTA (NTK-adaptive wins on its target failure); its value is failure-agnosticism, bounded actions, and rollback.
- **Theory.** The tangent-rank bound is exact; predicting the covariance participation ratio from task and optimizer remains open. The time-dependent PINN PR (1.3–1.9, *below* the steady 2D value) is an intriguing anomaly — the training explored an effectively one-parameter family of states — worth a dedicated study.

- **External validity.** All experiments are 1–10 seeds per configuration on a single-GPU campaign; the preregistration is in-repo rather than third-party (OSF archival is planned).

9 Conclusion

Check superposition before you sparse-code. We measured, on both sides of the boundary, what the SAE literature assumes: the tool works exactly where the representation is genuinely high-rank — and PINN activations are curves, not clouds. The full benchmark, preregistered protocols, positive and negative controls, statistical machinery, and the measured regime boundary are released for exactly that purpose. **The operator causal-asymmetry thread T1 is closed at H16b:** the high-n battery (n=20, direction-reversal criterion) fires H16b — 6/16 Bonferroni survivors but 0/16 direction-reversal survivors; the operator causal asymmetry is not a boundary-crossing signal.

Reproducibility statement. All numbers in this paper are regenerated from committed JSON artifacts (runs/) by `scripts/generate_tables.py` and `generate_figures.sh`; a CI job (`scripts/check_results_grounded.py`) fails the build if any results document cites a number ungrounded in an artifact. 155 unit tests; Dockerfile pinned; seeds documented per run; full campaign ≈ 6 GPU-hours (RTX-class). Preregistrations were committed before each stage ran; outcomes are recorded exactly as the decision rules fired.

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